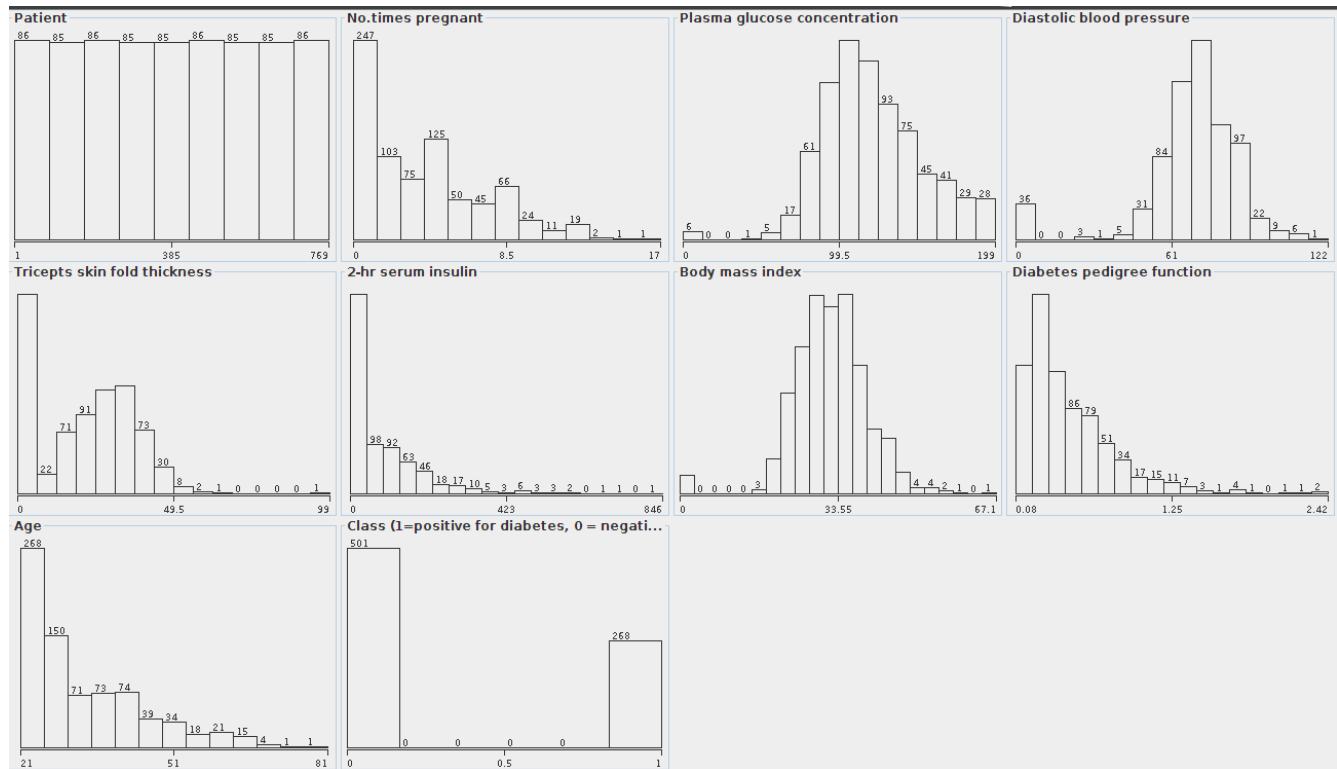


Missing values method with WEKA

Experimental Procedure

In this assignment I first load the dataset into WEKA tool using preprocessing panel.



I applied an unsupervised filtering technique by choosing “Replacemissingvalues” option. Then, my intent was to apply the C45(J48) Tree to analyze the dataset based on its attributes values. However, because part of the data was not nominal this was not possible now. Therefore, I proceed to apply the following trees available in WEKA. Later in this report we will see better results after I applied a filter for the those data that are nominal in the dataset.

=== Evaluation on training set ===					
=== Summary ===					
Decision Stump Tree		M5 Tree		REP Tree	
Correlation coefficient	0.4265	Correlation coefficient	0.5039	Correlation coefficient	0.484
Mean absolute error	0.3715	Mean absolute error	0.3385	Mean absolute error	0.322
Root mean squared error	0.431	Root mean squared error	0.4127	Root mean squared error	0.4206
Relative absolute error	81.8075	Relative absolute error	74.4391	Relative absolute error	70.8034
%		%		%	
Root relative squared error	90.4475	Root relative squared error	86.4687	Root relative squared error	88.1204
%		%		%	
Total Number of Instances	769	Total Number of Instances	769	Total Number of Instances	769

The next step was to apply the KNN computation on the dataset.

IB1 instance-based classifier
using 1 nearest neighbour(s) for classification

Correctly Classified Instances	507	65.9298 %
Incorrectly Classified Instances	262	34.0702 %
Kappa statistic	0.1706	
Mean absolute error	0.3856	
Root mean squared error	0.5199	
Relative absolute error	84.8961 %	
Root relative squared error	109.1069 %	
Total Number of Instances	769	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	ROC Area	Class
	0.848	0.694	0.696	0.848	0.764	0.627	0
	0.306	0.152	0.519	0.306	0.385	0.627	1
Weighted Avg.	0.659	0.505	0.634	0.659	0.632	0.627	

=== Confusion Matrix ===

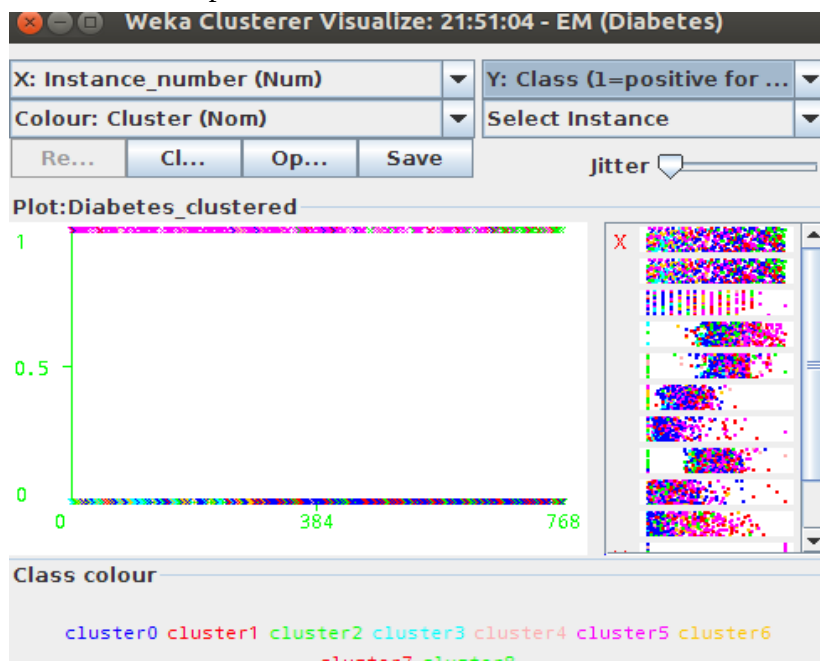
```

a  b  <-- classified as
425 76 |  a = 0
186 82 |  b = 1

```

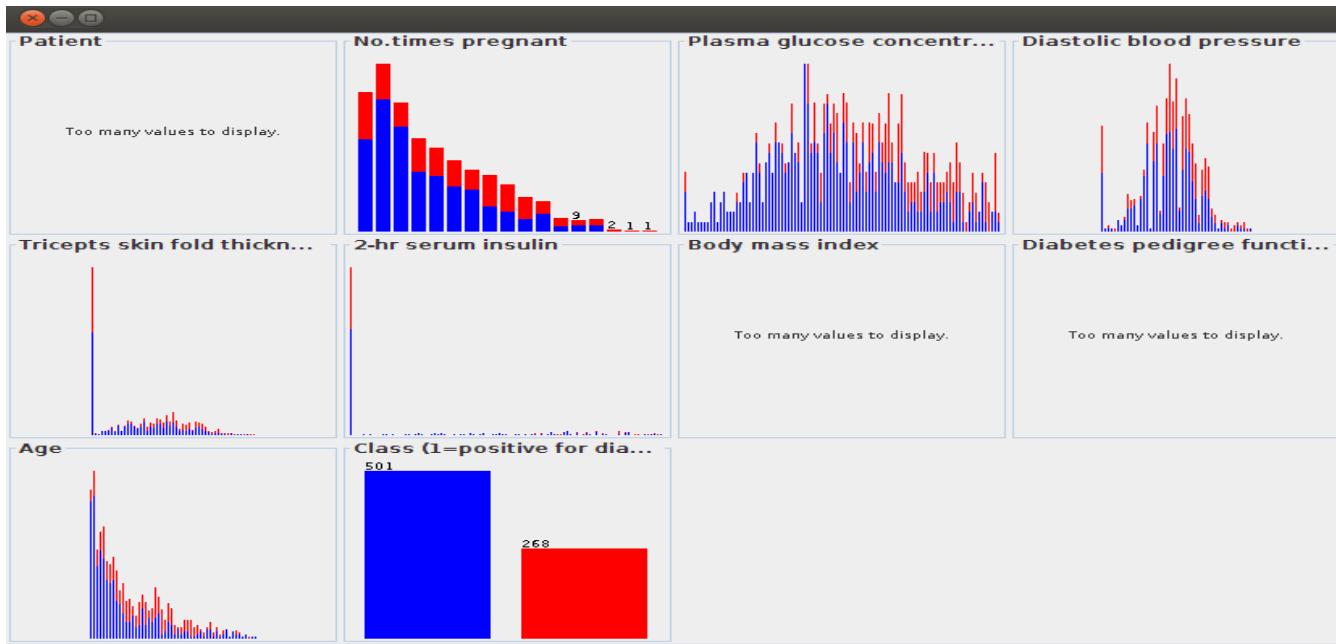
Below a visual representation of the KNN.

Clustered Instances



0	240 (31%)
1	32 (4%)
2	28 (4%)
3	45 (6%)
4	21 (3%)
5	195 (25%)
6	51 (7%)
7	88 (11%)
8	69 (9%)

Because I was not able to run the J48 algorithm on the dataset I decided to apply the “NumericToNominal” filter to be able to make use of this feature in WEKA. And, as we observe below, the results were better than those obtained in the first test.



On this dataset I apply filtering algorithm for litwise deletion. Then, in the classify panel we choose C4.5 Decision three classifier and start the analysis using 10 fold cross validation.

=== Classifier model (full training set) ===

J48 pruned tree

Number of Leaves : 1

Size of the tree : 1

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	501	65.1495 %
Incorrectly Classified Instances	268	34.8505 %
Kappa statistic	0	
Mean absolute error	0.4541	
Root mean squared error	0.4765	
Relative absolute error	99.9709 %	
Root relative squared error	100 %	
Total Number of Instances	769	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	ROC Area	Class
1	1	1	0.651	1	0.789	0.496	0
0	0	0	0	0	0	0.496	1
Weighted Avg.		0.651	0.651	0.424	0.651	0.514	0.496

=== Confusion Matrix ===

```
a  b  <-- classified as
501  0 | a = 0
268  0 | b = 1
```

Next step is to apply KNN computation on the dataset. Then, in the classify panel we choose IBK and start the analysis using 10 fold cross validation.

=== Classifier model (full training set) ===

IB1 instance-based classifier

using 1 nearest neighbour(s) for classification

=== Stratified cross-validation ===

=== Summary ===

Correctly Classified Instances	507	65.9298 %
Incorrectly Classified Instances	262	34.0702 %
Kappa statistic	0.1706	
Mean absolute error	0.3856	
Root mean squared error	0.5199	
Relative absolute error	84.8961 %	
Root relative squared error	109.1069 %	
Total Number of Instances	769	

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	ROC Area	Class
	0.848	0.694	0.696	0.848	0.764	0.627	0
	0.306	0.152	0.519	0.306	0.385	0.627	1
Weighted Avg.	0.659	0.505	0.634	0.659	0.632	0.627	

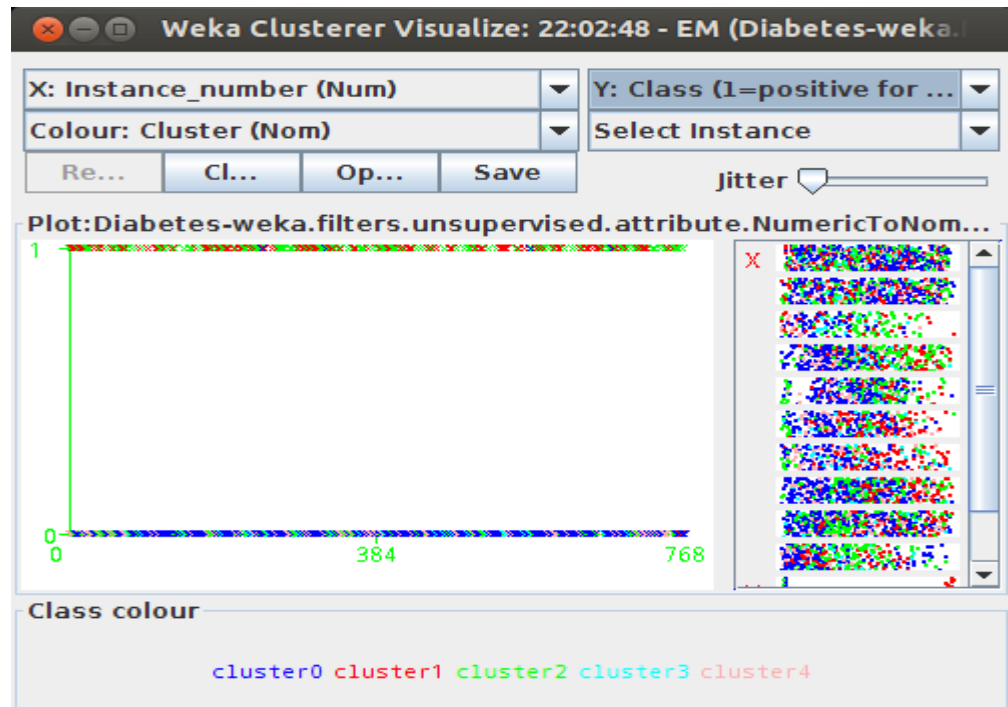
=== Confusion Matrix ===

```
a  b  <-- classified as
425 76 | a = 0
186 82 | b = 1
```

In this assignment the comparison has been done on the basis of accuracy/correctly classified instance and confusion matrix.

Clustered Instances

0	310 (40%)
1	128 (17%)
2	188 (24%)
3	46 (6%)
4	97 (13%)



Predicted Classes

Actual Classes	TP	FP
	FN	TN
	P	N

True Positive(TP) – These are the positive tuples that were correctly labeled by the classifier.

True Negative(TN) – These are the negative tuples that were correctly labeled by the classifier.

False Positive(FP) – These are negative tuples that were incorrectly labeled as positive

False Negative(FN) – These are the positive tuples that were mislabeled as negative

Accuracy: This is based on the probability that the test yields correct results, and it is calculated as = $(TP+TN)/(P+N)$ where, $P=TP+FN$ and $N=FP+TN$

Algorithm used C4.5 classifier	Correctly classified instances	Incorrectly classified instances	Accuracy	Mean absolute error
Litwise deletion	507	262	65.9	0.322
Mean/mode imputation	507	262	65.9	0.38
KNN(K-nearest neighbor) imputation	501	268	65.15	0.45

According to experimental results, correctly classified instances for Litwise deletion is about 507 and for mean/mode imputation is about 507 too. Correctly classified instances for KNN imputation is about 501, which is less than previous two algorithms. Accuracy of KNN is 65.15% which is also less than other two techniques so KNN is not good approach in this case to deal with missing data imputation as compared to other two techniques. Having said that, I believe that this is due to the fact that no missing data was found in this dataset and main in this dataset is due to its values authenticity and not to missing data.

CONCLUSION

Missing values are regarded as serious problem in most of the information system due to unavailability of data and must be impute before the dataset is used. Here the a diabetes dataset has no missing values and therefore our approach was not successful. To handle missing values three techniques were used named as Litwise deletion, mean/mode imputation, KNN (k nearest neighbor). Firstly dataset is loaded into WEKA tool. Apply these missing techniques individually on this dataset, which results as three imputed datasets. Then C4.5/J48 classification algorithm is applied to these replaced datasets and their results are compared in order to evaluate the efficiency and accuracy of missing data techniques on the basis of experimental results accuracy. KNN results were less than the other two techniques, as explained before the only explanation to this is that this dataset has no missing values. However, issues of its attributes values authenticity must be analyzed.

What are the differences?

Based on my results, in the first two techniques there was no difference. Then, when the KNN technique was applied there was a very small difference.

Did you improve the results?

Surprisingly, the results of KNN happened to be worst than that the previous two techniques. I have used KNN before, and it has given me very good results when replacing missing values and/or filtering them. However, in this case the results were bad because the dataset was not missing any value. The discrepancies in the dataset are not due to missing value, but to inconsistent data.